

NL-to-SQL Agent: Query Accuracy, Security & Safety Audit

1. Benchmark Testing Methodology

Executed 150 complex business questions across sales, inventory, and marketing schemas covering multi-table JOINS, GROUP BY aggregations, window functions, and nested queries.

2. Accuracy & Safety Evaluation

Query Complexity	Test Queries	Valid SQL Generated	Exact Data Match	SQL Injection Defended
Simple Filtering (Single Table)	50	100% (50/50)	98.0%	100%
Aggregations & Grouping	40	97.5% (39/40)	95.0%	100%
Multi-Table JOINS (2-4 Tables)	40	95.0% (38/40)	92.5%	100%
Adversarial Injection Attempts	20	0% (Refused)	N/A	100% (PASS)
Overall Performance	150	97.3%	94.7%	100% (PASS)